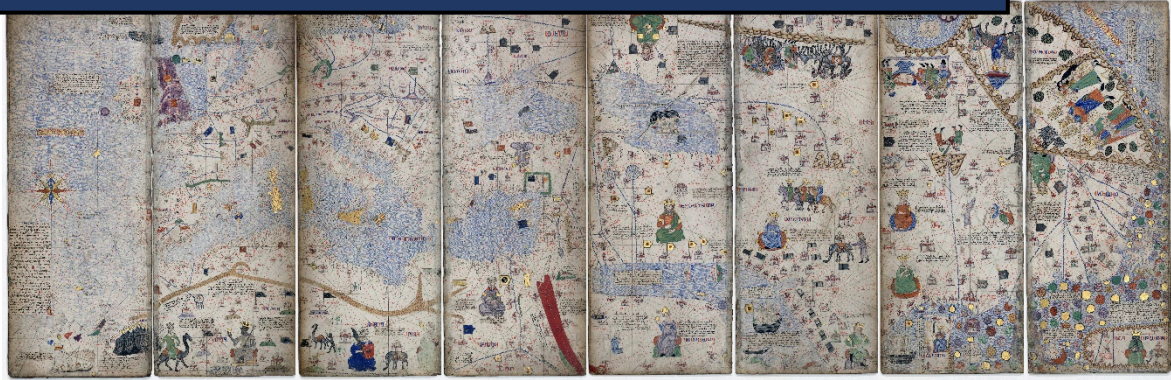


GEOGRAPHY, MOVEMENT, & THE PLACE OF REDEMPTION IN THE CATALAN ATLAS



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INTRODUCTION

A landmark in the history of mapmaking, the *Catalan Atlas* has long been recognized as an important expression of the socio-cultural aspects of travel. Commissioned by King Pedro IV of Aragon (1319–1387) and completed in 1375 by the Jewish scholar and cartographer Elisha Ben Abraham Cresques (1325–1387), the *Catalan Atlas* was given as a gift to King Charles V of France (1338–1380). It has remained in France since it was given to Charles V, eventually becoming part of the Bibliothèque nationale. The name *Catalan Atlas* comes from the use of Catalan language in the text across the work. This is representative of the language's spread throughout Aragon and the surrounding regions in the Middle Ages.¹

This monument was originally composed of six sheets of vellum each measuring 64.5 by 50 centimeters [Figs 1a-f]. At present, the *Catalan Atlas* exists in 12 sheets: in the sixteenth century, each sheet was cut in half and mounted on a wood panel, measuring 64.5 by 50 centimeters, resulting in 12 sheets of vellum mounted to the fronts and backs of seven panels (with the front of the first panel and back of the seventh panel left blank to act as covers for the foldable set).² On the first four oblong sheets of vellum are cosmological materials and on the other eight sheets is a map of the world. The resulting ensemble spans a total of 1.3 square meters.³ The first audience of the *Catalan Atlas* viewed it in its original form of six sheets, thus this paper will address the sheets in the same fashion, numbered 1-6. Intended to be viewed from

¹ Lluís Cabré, Alejandro Coroleu, Montserrat Ferrer, Josep Pujol, and Albert Lloret, *The Classical Tradition in Medieval Catalan, 1300-1500: Translation, Imitation, and Literacy* (Woodbridge: Boydell & Brewer Ltd, 2018), 21–24.

² "Spanish 30," Department of Manuscripts, National Library of France, Accessed June 8, 2024, <https://archivesetmanuscrits.bnf.fr/ark:/12148/cc78545v>.

³ "Spanish 30," <https://archivesetmanuscrits.bnf.fr/ark:/12148/cc78545v>.

above, the map would have been laid flat on a table so one could walk around it, enjoying a bird's-eye view of the earth and the ability to study it from all sides. For the purposes of this discussion, the orientation of the map positions the compass rose on the viewer's far left, with north at the top. Additionally, to distinguish the cosmological sheets from those of the map, I refer to the latter four sheets as the *Atlas's* portolan chart, as this is the genre of the map.

The *Atlas* has garnered scholarly attention for its geographical accuracy and for being the earliest medieval portolan chart to employ a compass rose.⁴ The opening two sheets contain cosmological texts and diagrams along with a zodiac wheel, and the remaining four sheets illustrate the world. Consistent with the genre of portolan charts, on each of the *Catalan Atlas's* sheets that make up the world map are 16 vectors arranged around a 17th central point with rhumb lines connecting each vector to the center as well as to the points adjacent and across the circle [Fig. 2]. These lines are useful not only for plotting courses of travel but also for visualizing connections across the whole of the map. Most important, the *Atlas* showcases a wealth of text and images that, when pieced together—and pieced together they must be—enact a complex theory of the world it depicts. These images and texts were understood to be reflective of lived realities, interactions, and exchanges between and among peoples from across the globe. Pieced together in a world picture, the *Catalan Atlas* grounded a theory of eternity within the ephemeral earth. This happened in the descriptions of cities and political territories as well as in details of meticulously laid out commercial routes, which became landmarks and signposts along the soul's journey through this life en route to the next one.

⁴ Clara Estow, "Mapping Central Europe: The Catalan Atlas and the European Imagination," *Mediterranean Studies* no. 13 (2004): 1–16, <https://www.jstor.org/stable/41166962>.

As a gift to the French king, the *Atlas* served as a reminder of Aragon's strength and stability for one of Pedro IV's closest allies. The *Atlas* also asserted Aragon's role as a microcosm of the world's place in the universe through the representation of the kingdom's political interests via its flag. For Pedro IV, commissioning the *Catalan Atlas* was a means to document his political achievements following a time of violence, unrest, and instability.⁵ The commissioning of a lavish, elaborate work such as the *Atlas* implied that, despite a decline in citizens' respect for public authority, the kingdom of Aragon wavered neither in the wake of the Black Death nor in the face of *la Guerra de los dos Pedros* (the War of the Two Peters), fought between the crowns of Aragon and Castile from 1356–1375.⁶ The strength and perceived prominence of the kingdom is marked on the *Atlas* in the same way it was marked on conquered lands—with the flag of Aragon. The red and gold leaf-striped flag appears multiple times across the map, on land and in bodies of water [Fig. 3]. The inclusion of the flag in its various locations was meant to represent Aragon's interests in the Mediterranean basin and to depict the kingdom as a hub for trade and commerce, while also framing it as a stable center of peaceful political exchange, thanks to the negotiations and policies of Pedro IV.⁷

Viewers & Viatores:

In creating this detailed, complex map that accounts for spiritual travel in addition to physical mobility, Cresques drew heavily from Isidore of Seville's *Etymologies* and Marco Polo's *Description of the World*, debts that are evident in many of the legends on the map. These sources, in addition to Claudius Ptolemy's *Geography* and Pliny the Elder's *Natural History*,

⁵ Joseph F. O'Callaghan, *A History of Medieval Spain*, (Ithaca and London: University of Cornell Press, 1975), 408.

⁶ O'Callaghan, *A History of Medieval Spain*, 408, 420–24.

⁷ Jocelyn N. Hillgarth, *The Spanish Kingdoms, 1, 1250-1410 — Precarious Balance* (Oxford: Oxford University Press, 1975), 233–271.

have been integral to understanding the *Atlas* and situating it within broader 14th-century intellectual and political developments. More recently, Katrin Kogman-Appel has described the *Atlas*'s significance as extending beyond the organization of spatial information to include ideas about identity and power, both of which revolve around the very notion of mobility.⁸ Additionally, Sandra Sáenz-López Pérez has discussed the eschatological imagery in the northeastern corner of the *Atlas* and debated the identities of the figures present there.⁹ Julio Samsó Moya and Juan Casanovas have focused on many aspects of the *Atlas*, including the diagrams on the cosmological pages and their use in calendar reckoning.¹⁰ Scholars such as Clara Estow and Sonja Brentjes have written on specific geographic regions on the *Atlas* and given attention, respectively, to the depiction of the European imagination and the illustrations rooted in Asian image traditions.¹¹ Building on such important work, I propose that the *Catalan Atlas*'s construction of mobility extends beyond court or nation to include that of body and spirit. In doing so, I suggest, the *Atlas* presents the earth as an ephemeral realm through which the wayfarer, or *viator*, must travel before arriving at the ultimate destination, the afterlife.

The Latin term *viator* would have referred to a wayfarer or traveler, especially one concerned with earthly travel while on the way to salvation.¹² This person could have been a

⁸ Katrin Kogman-Appel, *Catalan Maps and Jewish Books: The Intellectual Profile of Elisha Ben Abraham Cresques (1375-1387)* (Turnhout: Brepols, 2020), 24.

⁹ Sandra Sáenz-López Pérez, "La Representación de Gog y Magog La Imagen Del Anticristo En Las Cartas Náuticas Bejomedievales," *Archivo Español Del Arte* no. 78 (2005): 263–76, <https://doi.org/10.3989/aearte.2005.v78.i311.181>.

¹⁰ Julio Samsó Moya and Juan Casanovas, "Cosmografía, astrología y calendario," in *El Atlas catalán de Cresques Abraham* (Barcelona: Diáfora, 1975), 23–36.

¹¹ Estow, "Mapping Central Europe," 1–16. Sonja Brentjes, "Revisiting Catalan Portolan charts: Do they contain elements of Asian provenance?," in *The Journey of Maps and Images on the Silk Road*, ed. Philippe Forêt and Andreas Kaplony (Leiden and Boston: Brill, 2008), 181–201.

¹² Luigi Tomasi, "Homo Viator: From Pilgrimage to Religious Tourism via the Journey," in *From Medieval Pilgrimage to Religious Tourism: The Social and Cultural Economics of Piety*, ed. William H. Swatos, Jr. and Luigi Tomasi, *Religion in the Age of Transformation* (Westport, Connecticut and London: Prager Publishers, 2002), 1.

messenger, an itinerant worker, one who guided the way for others, or a user of the *Catalan Atlas*. Unlike someone who may have been travelling out of necessity, the *viator* was said to act on a deep desire to know and was perhaps even driven by a sense of longing. The desire to travel placed the *viator* in a position of constant searching—searching for meaning, searching for the sacred, searching for more. For the purposes of this paper, the term *viator* will refer to users of the *Catalan Atlas* who travelled only across the surface of the map, making theoretical and non-linear journeys corresponding to the ideal of autotelic movement, as this is how the *Atlas* seemed to have been designed to be used. These users would have been members of the Aragonese and French courts and would have brought their Christian beliefs to their understandings of the *Atlas* and its contents. The *Atlas*'s portolan chart pictures the earth in order to show what was possible and to facilitate travel by proxy as the *viator* imagined their body taking the journeys their eyes traced across the *Atlas*'s sheets. These journeys could have been economic or political, but they also could have been exploratory or eschatological. Most important, they were imaginative, participatory, and dependent upon the perspective and beliefs of the *viator*.

Medieval travel and pilgrimages, according to Luigi Tomasi, were a search for knowledge just as much, if not more, than they were a desire to relocate oneself.¹³ Pilgrimages in particular were about the processes of the journey and the sanctification of the soul as one moved toward their destination. Pilgrims, broadly speaking, were those travelling in pursuit of religious experiences, some for the purposes of gaining salvation, others for a closer encounter with the divine, and still others seeking forgiveness, healing, or blessings.¹⁴ A type of pilgrim, *viatores* met these criteria as well, but were differentiated in the way that the physical destination

¹³ Tomasi, "Homo Viator: From Pilgrimage to Religious Tourism via the Journey," 1–3.

¹⁴ Tomasi, "Homo Viator: From Pilgrimage to Religious Tourism via the Journey," 3–4.

of their quests were not always marked at the outset of their journey. The *viator*'s destination was still specific—eternity in heaven through salvation—albeit less concrete. While pilgrims began with a destination in mind, Jerusalem, for example, a *viator* embarked on a journey in pursuit of knowledge and the divine, not necessarily knowing where it might take them. With their gaze fixed ahead, pilgrims and *viatores* alike focused on the long trajectory toward redemption. Along the way, the journey—perhaps best thought about as a quest for understanding the divine order they were already participating in—reminded the *viator* that the reception of grace is the only path to the place of redemption. Framing the soul's journey through life as a metaphor for earthly travel, the *Catalan Atlas* invited the *viator* to ponder their place in the divine order. Through the intertwining of vectors with illustrations and informational texts, and with the assumption that the user would bring their own beliefs to their interpretations, the *Catalan Atlas* encouraged the *viator* to consider their soul's trajectory by journeying across its surface and to pause now and then to weigh the implications of particular sites or concepts.

The variety of those implications shone through, for instance, in the inclusion of familiar biblical sites (e.g., Noah's Ark at Mount Ararat), places of radical uncertainty (e.g. a city abandoned due to an onslaught of snakes), and even sites of no particular economic or political importance beyond apparent novelty (e.g. a city where ships boast eight or ten masts with reed sails). Alongside these sites and routes are depictions of pearl divers, elephants, notable rulers, camels, horses, and cremation practices – all accompanied by descriptive legends. The legends include both utilitarian information typical of portolan charts, such as names of cities, descriptions of resources, and other practical pieces of data. They also include non-utilitarian and ethnographic tidbits, such as observations about a population's physical stature, the types of clothing they wear, fishing rituals, and so forth. These legends even include warnings of

potential dangers and threats, concerns about accuracy, declarations of what was considered to be fact, and descriptions of ideas and practices surrounding death.

Drawing on those legends, as well as on the map's visual and physical characteristics and on its economic and political implications, this paper will address the *Catalan Atlas's* philosophical, religious, and environmental significance. With particular attention to the metaphorical and eschatological implications of travel, I propose that the *Atlas* established parallelisms among body, soul, earth, and cosmos to thematize earthly movement as a microcosm of the one's trajectory toward either redemption or perdition. The microcosm I have in mind is less one of simple mirroring and more of enacting virtual movements across the *Atlas*. I argue that these virtual movements simulate physical travel across the earth, which for the virtual traveler, became a metaphor for the soul's movements through earthly life toward the afterlife. Moving through the map, beginning with the cosmological sheets and ending in the Caspian Mountains, this paper will explore the ways viewers, cast as metaphorical *viatores*, might have used the *Catalan Atlas* to conceptualize themselves within the broader context of the universe and to reconcile their expectations of the afterlife with their time on earth.

As mechanisms of meaning but not necessarily means to an end, medieval maps offer viewers ways to construct trajectories of agency and patterns of movement. The materials offered by the Catalan Atlas include illustrations alongside descriptions of places, a compass rose, and rhumb lines—all of which the *viator* used to construct their trajectory. As the cartographer tasked with mapping the end, Cresques offered the *viator* images of figures who could be identified but the interpretation of those figures and the legends accompanying them was left up to the individual viewing them. In turn, with images as the starting point for visual exploration, the conceptualization of voyaging became about the construction of imaginary worlds and the

making of mental pathways. J.B. Harley describes maps as mediating between the mental inner world and the outer physical realm, making them “fundamental tools for helping the human mind make sense of its universe at various scales.”¹⁵ In a practical sense as well as a spiritual one, the *Catalan Atlas* guided the *viator* from point to point, through areas of transition, while highlighting indeterminate and inconclusive forms of movement. Through the intertwining of illustrations with textual information, the *Atlas* extended an open invitation to journey across its surface, with pauses at points that highlight the varieties and consequences of travel. Including bits of information that seemingly served no distinctly utilitarian purposes reinforced the idea that this object was created to do more than convey directions from one location to another. The map reminded the *viator* that the mobility of the soul cannot be separated from earthly travel and movement, nor can it be disentangled from the very stuff of the earth, from the rocks, trees, water, and air.

CONCEPTUALIZING & MAKING SENSE OF THE UNIVERSE: COSMOLOGICAL SHEETS & HIBERNIA

The first two of the *Atlas*'s six sheets set the stage for the following four by establishing the framework the depicted world exists within [Fig. 1a and 1b]. This framework reflects the cultural landscape of Aragon its merging of Christian, Islamic, and Jewish calendrical, astronomical, and astrological forms of knowledge. This is significant as the next hundred years would be characterized by forced conversions to Christianity, leading up to the Inquisition in the late 1470s. Thus, while the portolan chart portion of the *Atlas* is permeated with themes of Christian conquest, Cresques used its framing material to situate the portolan chart within Aragon's broader intellectual landscape, leveraging the commission by the Christian king as an

¹⁵ J.B. Harley, “The Map and the Development of the History of Cartography,” in *The History of Cartography* ed, J.B. Harley and David Woodward, (Chicago, University of Chicago Press, 1987), 1:1.

esoteric platform that held insights from Christians, Muslims, and Jews about the construction of the calendar and the knowledge held by the stars in the skies. For the patron, this reflected the Christian courtiers considerations of themselves as be broad-minded while holding fast to their belief in Christianity as the only true faith.¹⁶ Thus, in a map such as the *Catalan Atlas*, what is produced by the cartographer is not an image of the world as it is, but an image of the world as the patron, and sometimes the maker, understood it. In the case of the *Catalan Atlas*, for its Christian *viatores*, the image of the world was also an image of the divine order and the ways in which the Christian ought to move through the world as a recipient of the grace that promises redemption and leads to everlasting life.

The opening sheet of the *Atlas* is divided into four columns that set up the cosmological framework for the four sheets that make up the *Atlas*'s portolan chart. The first, third, and fourth (reading from left to right) are filled entirely with text detailing the construction of the Christian calendar, information that was commonly included in other manuscripts contemporary with the *Catalan Atlas*. In the first column is a lunary containing astrological information that was popular throughout Europe, including prognostications as well as recommendations for medical treatment keyed to each day of the lunar month.¹⁷ The third and fourth columns contain a Catalan translation of an excerpt of Book One of Honorius Augustodunensis's *Imago mundi* (c. 1110). The translation, which evolves into a paraphrase of the text toward the end of the fourth column, provides information on the tides and for determining the local time based on the position of the Great Bear constellation relative to the North Star.¹⁸

¹⁶ O'Callaghan, *A History of Medieval Spain*, 502–506.

¹⁷ Kogman-Appel notes that these recommendations follow the conventions for the 28 lunar mansions in Islamic astronomy traditions. Kogman-Appel, *Catalan Maps and Jewish Books*, 77, 84.

¹⁸ Kogman-Appel, *Catalan Maps and Jewish Books*, 77–78.

The second column shows three diagrams that illustrate the texts flanking them. The two circular diagrams illustrate the construction of the calendar, while the diagram of the human body depicts how the calendar ought to be used. The top diagram, a blue circle set against a red background, is a tidal *rota* set within a wind rose. This was used to determine the course of tides. Just below it is a diagram that resembles an astrolabe. Associated with calendar reckoning, the device depicted in the diagram was used to determine the Christian holy days.¹⁹ The third diagram on this column is a zodiac man. The text on and surrounding the body shows how the zodiac signs correspond to specific parts of the anatomy and gives instructions for healing in accordance with the calendar, dovetailing with the text in the first column.

The second cosmological sheet is given over to an astrological diagram similar to late Antique synagogue floor mosaics. The diagram outlines the cosmological framework of the universe, depicting the planets and the phases of the moon. In the corners of the diagram, just outside the boundaries of the circular diagram, are four figures, each holding a banner and oriented toward the center of the sheet. These figures represent the four seasons, a tradition consistent with synagogue floor mosaics from across the Mediterranean and North Africa [Fig. 4].²⁰

At the center of the diagram is a small and badly damaged illustration of a man studying the skies [Fig. 5]. Kogman-Appel has plausibly suggested that this image depicts “at the center of the universe [sic] the individual scholar trying to make sense of the world. Study and scholarship lead to knowledge about the world and the cosmos and, in some sense, seem to

¹⁹ Kogman-Appel, *Catalan Maps and Jewish Books*, 78. Based on Julio Samsó Moya and Juan Casanovas, “Cosmografía, astrología y calendario,” 26.

²⁰ Lee I. Levine, “The Zodiac and Other Greco-Roman Motifs,” in *Visual Judaism in Late Antiquity: Historical Contexts of Jewish Art* (New Haven and London: Yale University Press, 2012), 322–23.

dominate the earth.”²¹ Throughout the *Atlas*, and particularly in this image, Cresques captured the human desire to understand the world and one’s place in it. That desire, for the Christian *viator*, is perhaps better understood as an effort to gain some small sense of divine order and of how best to participate in it. That order set the portolan chart within the cosmological framework laid out on the *Atlas*’s opening two sheets, and on the following four sheets we see how the order continually oriented the *viator*’s journey toward the afterlife.

As the *viator* moved from the framing material to the first sheet of the *Catalan Atlas*’s portolan chart, they were met with a compass rose, reminiscent of the center of the tidal *rota* at the top of the first cosmological sheet. On the far-left side of the *Atlas*’s portolan chart, just above the compass rose and next to the island labeled Irlanda, is a legend describing the islands of Hibernia.²² Laden with details that are not easily mappable, Hibernia immediately presented the *viator* with eschatological concerns in its description of the islands as a place that people leave before dying [Fig. 6]. According to the legend, on one of the Hibernian islands people were transported away before dying. On another island, before one gave birth, they were also transported away. Where they went, how they got there, and (in the latter case) whether or not the parents and their offspring were able to return are all questions left unanswered by the *Atlas*.

²¹ Kogman-Appel, *Catalan Maps and Jewish Books*, 86–87.

²² “In Hibernia [Ireland] there are many wonderful islands whose existence can be credible; among them, there is a small one where men never die, because when they are about to die of old age, they are transported outside the island. There are no snakes, frogs, nor poisonous spiders because the soil repels them given that this is where Lacerie Island [Cléire/Clear Island] is located. Furthermore, there are trees that attract birds like ripe figs. There is also another island where women never give birth because when they are about to give birth, they are taken outside the island as it is customary.” Juan Ceva, “Panel III,” *Catalan Atlas Legends* (web page), *The Cresques Project*, March 18, 2014, <https://www.cresquesproject.net/catalan-atlas-legends/panel-iii>.

All the *viator* saw is a description of the place from which they departed on the cusp of either the beginning or the end of earthly life.

The question of where the Hibernian people went before they die was no less easy to answer than the question of precisely what happens *when* one dies. The act of transporting from one life to the next presented a venture through the truest unknown, a journey through a placeless space and into the promised afterlife. People were departing from their homes, families, and communities at critical life moments such as death and childbirth, and these life moments themselves were framed as departures from the realm of ordinary daily living. Because the Hibernia legend does not specify where those who depart go, it left arrival an open question. This, in turn, placed an accent on the fact of travel itself rather than on its starting and ending points. As a result, the *Catalan Atlas* brought forward the metaphysical implications of such travel. Combined with their socio-culturally determinate endpoints and the Christian beliefs of the *Atlas*'s primary audience – Aragonese and French viewers undoubtedly thought of themselves as *knowing* what birth and death were – these indeterminate journeys reminded the *viator* that travel was not only about moving from one place to another but was about the very notion of mobility itself because the soul was in motion just as much as the body.

Confronting those enigmatic Hibernian journeys, the *viator* was reminded that the heavens and the soul's journey toward them could not be separated from earthly travel and movement. In instances of such radical uncertainty – indeterminate departures on the eve of determinate events – the *viator* was permeated by a sense of travel that focuses our attention on movement itself. And yet, the theme is that of moving those who are already in motion: the quick departed one location on the eve of becoming the dead; the expectant mother departed another location after the quickening of the fetus but before its arrival. Here, Hibernia echoed divine

order, both by framing death as a departure from the realm of the living and a rebirth into the afterlife and by paralleling death with childbirth, itself a departure from the womb into the realm of the living. The legends for these two sites thus raised questions specifically not to answer them and relied on the Christian *viator* bringing their own beliefs to their reading of the map in order to focus the *viator's* attention on the promise of redemption in the afterlife.

KNOWLEDGE IS POWER: THE WAYS OF WATER

In addition to being the first portolan chart to employ a compass rose, the *Catalan Atlas* is also unique in its use of visual means to identify bodies of water.²³ Across the *Atlas*, wavy blue lines mark the locations of the oceans and seas, and thicker dark blue lines denote rivers. Such details are absent from earlier and contemporary portolan charts. The presence of water across the *Catalan Atlas* is significant for three main reasons. First, in a practical sense, and in tandem with the compass rose, illustrating water simplified the plotting of courses of travel by making it easier to visualize the positions of landmasses in relation to one another. Second, Cresques used the water as a visual field on which to include pieces of information and warnings about the things one might encounter in the seas—for instance, the locations of precious stones and gems as well as the potential home of sirens [Fig. 7]. Third and finally, beyond indications of locations, water was also used as a means to call to mind biblical narratives such as the exodus from Egypt through the Red Sea and the story of Noah and the ark. In each of these instances, the *Atlas* equipped the *viator* with insights, some new and others familiar, and implies the message that while knowledge is power, so too is water. In this section we will discuss the presence of water not only as a cartographic advancement, but also as an indicator of the divine order and the *viator's* participation in it.

²³ Kogman-Appel, *Catalan Maps and Jewish Books*, 177.

We see this power on display in a practical sense, as knowledge of the ways of water in a specific area grants access to the land. To organize and display one's knowledge of water on a map, was a demonstration of the flourishing intellectual activity in Aragon and an expression of Cresques's particular cartographic prowess.²⁴ The heavy use of blue pigment across the *Atlas's* portolan chart and its framing material, along with gold leaf, boasted the wealth of physical materials Cresques had at his disposal to illustrate the *Atlas*, and the materiality of the blue expanses would not have been lost on viewers. The depiction of water as a demonstration of knowledge, however, was arguably more important. The ability to map shorelines and place islands within bodies of water elevated the caliber of portolan charts and set a new standard for the appearance of royally commissioned maps by sending the message to viewers of the *Atlas* that Aragon was the leader in cartographic advancement. Alongside the depictions of water, the *Atlas's* legends included important pieces of information about navigating particular areas. For instance, the seas near Sweden are described as being frozen six months of the year, making travel all but impossible.²⁵ Indicating where the sea was too rough to sail across preserves the lives of sailors and informed decisions about where to set up ports. Control over ports meant control over the things and people that enter the territory. In addition to showcasing knowledge of the seas, lakes, and rivers, the visual indication of water on the *Atlas* also marked the edges of the explored world, demonstrating knowledge of current affairs in discovery and travel alongside cartographic advancement.

²⁴ Standard practice for portolan charts prior to the creation of the *Catalan Atlas* was to assume the presence of water rather than to illustrate it in blue pigment.

²⁵ "This sea is called Sea of La Mancha, Sea of Gotilandia [Gotland] and Sea of Susia [Sweden]. Let it be known that this sea is frozen during six months of the year; that is to say, from mid-October until mid-March; and it is so hard that one can cross on top of it with an oxcart. The rough weather is due to the cold North wind." Juan Ceva, "Panel IV," *Catalan Atlas Legends* (web page), *The Cresques Project*, March 18, 2014, <https://www.cresquesproject.net/catalan-atlas-legends/panel-iv>.

Functionally, portolan charts were used to track the boundaries of the known world and to measure the progress of exploration with specific regard to harbors and ports. Portolan charts also had a commitment to accuracy that set them apart from earlier European medieval world maps which tended to be more conceptual in their representations of geographic space. The *Catalan Atlas* added a layer of complexity to the idea of that “known world” by allowing space for the unknown and the poorly understood; the location of a body of water might be established, but what lay beneath its surface remained a mystery. We see this in particular in the seas of the Pacific Ocean (present-day East China Sea, South China Sea, and Philippine Sea), which dominate the final sheet of the *Catalan Atlas*. On the first three sheets, land surrounds the waters, speaking to the importance of territorial claims for the map's Aragonese patron. Pedro's lands as well as those of his fellow kings and princes teem with riches, thrived through trade, and constantly jockeyed for supremacy. On the final sheet, however, the Pacific Ocean itself now teemed—with exotic islands, strange creatures, and untold wealth.

In the sea near Sri Lanka, for instance, mysteries abound. The legend mentions three types of sirens, but only two types are described, and only one siren is illustrated. One of those types is half fish and half woman, the other half bird and half woman.²⁶ Loosely following the description written by Isidore of Seville, the depicted siren appears to fall into the first category, having the upper body of a human woman and holding one tail in each hand, one red and the other violet.²⁷ Just above the siren, Cresques listed the types of fish and precious stones that

²⁶ “Sea of the islands of the Indies where one finds spices. Many ships from many nations cross this sea. Here, one finds three kids of fish named sirens: one is half fish, half woman, the other is half woman, half bird.” Ceva, “Panel VI,” *Catalan Atlas Legends* (web page), *The Cresques Project*, March 18, 2014, <https://www.cresquesproject.net/catalan-atlas-legends/panel-vi>.

²⁷ “People imagine three sirens who were part maidens, part birds, having wings and talons; one of them would make music with her voice, the second with a flute, and the third with a lyre. They would draw sailors, enticed by song, into shipwreck.” Isidore of Seville, *The Etymologies*, trans. Stephen A. Barney et al., Third (Cambridge: Cambridge University Press, 2011), 245, 322–326.

might be found in certain areas, based on the writings of Marco Polo and Isidore of Seville about the Pacific Ocean and the islands in it.²⁸ For the *viator*, travelling by proxy from the comfort of the court, the Pacific Ocean was presented as if an almost incomprehensible treasure trove on the farthest shores at the very ends of the earth. As such, that region served as a reminder that earthly ends or boundaries are illusory. Instead, they were new beginnings.

These new beginnings echo the eschatological concerns presented in Hibernia. The boundaries of the earth and life on it crash against the shores of the unexplored. Similar to the questions raised in Hibernia—questions that go unanswered in the earthly realm but ultimately point to the promised afterlife—the Pacific Ocean asked the *viator* to consider possibilities, namely sirens and treasures, that have yet to be discovered in order to conceptualize the parts of the world no one had seen. The presence of sirens indicated that the *viator* should proceed with caution in pursuit of the riches held within the Ocean's water, lest they fall into temptation. This paralleled the understanding of how the Christian ought to make their way through the world—in pursuit of righteousness and avoiding sin because true wealth is stored in heaven. For the Christian *viator*, the unknowns of the Pacific Ocean correlate to the anticipated transition from the hardships and temptations of earthly life into the glorious riches that await in everlasting life.

Promises of a life beyond (and yet always somehow dependent on) the court are visible to the *viator* throughout the *Atlas*, not only in the blue waves of the Pacific Ocean, but also in images that called to mind narratives of water. Shifting our focus to the center of the *Atlas*, we find two biblical instances of water symbolizing divine power. Catastrophic in both absence and

²⁸ Marco Polo, *Description of the World*, trans. Sharon Kinoshita (Indianapolis and Cambridge: Hackett Publishing Company, Inc., 2016), 147. Isidore of Seville, *The Etymologies*, 286.

excess, water is a fundamental component of the world as well as a precondition of life.²⁹ In the opening verses of the book of Genesis, water was present from the moment heaven and earth are created, before even light was spoken into existence.³⁰ Further on in the narrative, the earth was destroyed with an overabundance of water – an event referenced by Noah’s Ark at Mount Ararat [Fig. 8]. Isidore of Seville understood floods as acts of nature that had to be taken seriously, both in terms of the physical damage they could do, and in terms of what they might portend. He wrote: “one must bear in mind... that when rivers rise higher than normal, they not only bring destruction in the present moment, but they also signify something yet to happen.”³¹ The placement of Noah’s Ark next to a river attested to this, serving as a reminder that destruction is not the end, but is instead a new beginning. The inclusion of Noah’s Ark on the *Catalan Atlas* is significant because, much like the journeys presented in Hibernia, the journey Noah and his family embarked on was one of unknowns intertwined with promises of redemption. We see divine favor shine down on Noah’s family in the way they are spared from being consumed by the flood and again when a dove was sent out from the boat and returned with an olive branch, which, along with the rainbow in the sky, was taken a sign of safety, an indicator to begin again.

The second instance of water as a symbol of divine power occurred at the Red Sea [Fig. 9]. Similar to the story of Noah and the Ark, the Exodus narrative and in particular its events that take place at the Red Sea were marked by an act of re-creation that is redemptive and restorative. In both instances, an act of divine intervention destroyed the sinful and the oppressors with water while the faithful were spared. In the case of Noah and the Ark the floodwaters destroyed

²⁹ Steve Mentz, “Ice/Water/Vapor,” in *The Cambridge Companion to Environmental Humanities*, ed. Jeffrey Jerome Cohen and Stephanie Foote, Cambridge Companions to Literature (Cambridge: Cambridge University Press, 2021), 185–98.

³⁰ Genesis 1:2 (NRSV) “...and darkness was upon the face of the deep; and the spirit of God moved over the waters.”

³¹ Isidore of Seville, *The Etymologies*, 283.

everything on the earth's surface, with Noah and his Ark spared and floating atop the waters. Meanwhile, the Red Sea is presented as a passageway, both in the biblical narrative and on the routes of the *Catalan Atlas*. In the biblical narrative, the Red Sea loomed large and cast a shadow that threatened to end the Israelites' journey out of oppression before it began by washing away the precipice of liberation from slavery in Egypt. Yet, without so much as a boat, the Israelites safely crossed the Sea, leaving their captors to meet their demise in the Sea's wake.³² Viewed in parallel, the *Catalan Atlas*'s illustrations of Noah's Ark and the Red Sea called to mind narratives that functioned as representations of the soul's journey out of earthly oppression and toward the re-creation promised in the life to come.

Looking at the image of the Red Sea and its accompanying legend, the *Catalan Atlas* identifies it as a place through which spices pass from India to Alexandria. The body of water, easily identifiable by its red hue, extends from the lower right corner of the *Atlas*'s fourth sheet and contains one of the rhumb line convergence points marked in Figure 2. Along either side of the Sea are illustrations of eight cityscapes, four on each side. The legend at the top of the Sea indicates that nearby is a gorge that "the sons of Israel passed through" as they fled Egypt, and the legend accompanying the nearby mountains informs us that the region was a passageway along the route for many pilgrims "wishing to go to Mecca to see the arch of Mohamed that contains their law."³³ Framing the Sea and its surrounding areas as a passageway, the *Atlas* used the body of water as a point of convergence where different journeys and trajectories intersected with one another before continuing on their own paths, bringing us back to the motif of movement. From the Israelites' journey through the Red Sea to the late medieval pilgrim's route

³² Exodus 14:21–29 (NRSV).

³³ Ceva, "Panel IV," <https://www.cresquesproject.net/catalan-atlas-legends/panel-vi>.

around it, and even to a more mundane activity like the transportation of spices across it, the depiction of the Sea as a passageway reminded *viator* of the importance of the sites along the route to eternity. While the *viator* viewing the *Catalan Atlas* from the comfort of the court would not have embarked on any of the aforementioned journeys, they were still reminded to seek the divine in everything and every place.³⁴ In this sense, the places passed through become part of the destination as the *viator*, confident in their salvation, made their way through earthly life in pursuit of righteousness while awaiting redemption.

SAND & STONE: FROM THE LOP DESERT TO THE CASPIAN MOUNTAINS

As the *viator* journeyed across the *Catalan Atlas*, they were presented with tales of quests for resources alongside questions of life, death, and categories in between. Throughout the *Atlas*'s portolan chart, the legends denote the regions rich in spices, precious metals and stones, leather, various types of fruit, and exotic animals such as elephants. Other legends and illustrations detail processes of cremation, sati, and leaving islands before dying, as we saw in Hibernia. Also present are descriptions of the environmental conditions a traveler might encounter such as the regions in the north where the sea is frozen for six months of the year and the skies are dark even during the day. Pieces of information such as these reminded the *viator* of the cyclical nature of life on earth. Seasons of planting and growing give way to the harvest; as one life comes to an end, somewhere else a new life is beginning. The harsh northern winters eventually turn into spring, only to come back again the next year. In identifications such as this, the *Catalan Atlas* reminded the *viator* that everything on earth is ephemeral—even destinations,

³⁴ This goes back to Tomasi's discussion of the *homo viator* as one who is *constantly* in search of knowledge, seeking to know both "mundane reality and celestial mystery." Tomasi, "Homo Viator: From Pilgrimage to Religious Tourism via the Journey," 1.

which one ultimately will depart for the afterlife.³⁵ With that in mind, we now turn to two final places on the *Catalan Atlas* that frame both the earth and an individual's earthly life as but small parts of divine order. First, we will examine the Lop Desert as a motif of an evolving earth. After that, we will travel to eschatological images of the Caspian Mountains.

The Lop Desert, located in the Tarim river basin of northwest China, presented travelers with topographical mystery as well as significant challenges to survival. The desert's legend warns of the dangers one will face in attempting to cross it [Fig. 10]. It reads:

Let it be known that those that wish to cross this desert stop and rest during a week in a city named Lop. Here, expeditions and their animals relax/enjoy themselves. After that, they procure what is needed for the next seven months of their journey, because in the desert one travels an entire day and night before reaching potable water; however, every day and a half, they can find plenty of it, enough for fifty or a hundred people or even more. And if it happens that a rider, tired by the journey, falls asleep or for any reason he separates from his companions, he will often hear the voices of devils, similar to the voices of his companions, often calling him by his own name. In this way, the devils take him through the desert to and fro such that the traveler cannot find his own companions. A thousand stories are known about this desert.³⁶

The *Catalan Atlas* framed the crossing of this desert as a respite, but with an edge, a dark underside. Known for its harsh climate, the desert claimed the lives of many who attempted to cross it. Later study of the Tarim revealed that the almost perfectly flat terrain had three slightly depressed areas that became lakes when filled with water. These lakes, however, were not fixed in their locations. Due to severe sandstorms and the shallow depressions, as the Tarim River

³⁵ In his book *Stone: An Ecology of the Inhuman*, Jeffery Jerome Cohen uses the term "geophilia" to describe the space of such cycles, defining it as "a middle region of creation and innovation, a space of convivial wayfaring—experimenting, working, and living together, a place of differences and disorienting danger, a forging of alliance and embrace that gathers a world so vast that even stones become fellow travelers along the epochal, uncertain but never unaccompanied ways." Jeffery Jerome Cohen, *Stone: An Ecology of the Inhuman* (Minneapolis: University of Minnesota Press, 2015), 252.

³⁶ Juan Ceva, "Panel V," *Catalan Atlas Legends* (web page), *The Cresques Project*, March 18, 2014, <https://www.cresquesproject.net/catalan-atlas-legends/panel-v>.

changes course, so too do the locations of the terminal lakes.³⁷ While this information about the Tarim River was not known by Cresques, the *Catalan Atlas*'s description of the Lop Desert alludes to the changing terrain that had not yet been explained by scientific study. In the 14th century, when this information was still unknown, this meant that a consistent route across the desert could not be mapped. The traveler must be privy to the changing locations of fresh water as well as to the threat of deception by familiarity lest they lose their companions. Staying alert and in community on the journey, however, offered a sense of security and safety from being led astray while navigating the ever-evolving sandy expanse. While the Aragonese and French *viatores* were unlikely to find themselves crossing the Lop Desert, its presence on the *Catalan Atlas* again directed their focus back to the motif of movement. For the *viator*, moving through the desert virtually, this imagined journey pointed to an understanding of the earth being in motion in addition to facilitating the movements of those travelers who did personally experience the desert.

This understanding of an earth in motion manifested itself in the changing course of the Tarim River and its terminal lakes and is indicative of how deserts in general are shaped by cycles of transition. The Lop Desert on the *Catalan Atlas* emphasized the fact the neither the earth nor the *viator* were ever still but were, instead, constantly in motion, moving through continually changing circumstances. The description of the Lop Desert exemplifies this in how it presents a notion of being lost or wandering that is evocative of biblical narratives, similarly to the illustration of the Red Sea. Here, the environment can be understood as a prompt to memory in the way the physical body interacts with the physical earth, urging their spirit to “touch a

³⁷ Zhao Songqiao and Xia Xuncheng, “Evolution of the Lop Desert and the Lop Nor,” *The Geographical Journal* 150, no. 3 (1984): 311–21, <https://doi.org/10.2307/634326>.

world possessed of long futurity and deep past, a spatial expanse that stretches from the subterranean to the cosmic verge,” to quote Jeffery Jerome Cohen in his book *Stone: An Ecology of the Inhuman*.³⁸ Absent of waterways and greenery, deserts figure as threatening amorphous realms through which one must necessarily wander, one's steps dogged not only by thirst and hunger but also by disorientation. Recognizing this calls to mind the story of the Israelites, who, although they had a clear destination, were destined to wander before finally reaching their terminus. They, like the late-medieval pilgrim, began at a distinct point of origin and had a specific destination in mind even as they accepted the vicissitudes of travel through an unpredictable and often inhospitable world. The *viator*, too, moved through the unpredictable world toward a specific destination, but not one that could be placed on a map. In pursuit of eternity in heaven, the *viator* journeyed through the earthly realm with their gaze beyond the subterranean, focused on the promises of redemption.

Shifting from sand to stone, we now turn to the Caspian Mountains and the eschatological imagery held within them that points to the anticipated end of the earthly realm. While the miniscule pieces of the past contained within the Lop Desert's grains of sand are constantly being rearranged and reorganized by wind, waves, and human and animal activity, the stones of the Caspian Mountains keep time similarly to trees and the rings of their trunks—established, unmoving, and slow-growing. Formed more than 25 million ago, the Caspian Mountains stretch between the Black and Caspian Seas and mark the intersection of Europe and Asia.³⁹ On the *Catalan Atlas*, however, the Caspian Mountains sit directly above India and Sri Lanka, the waves of the Pacific Ocean crashing against them [Fig. 11]. Focused less on the motif

³⁸ Cohen, *Stone: An Ecology of the Inhuman*, 33.

³⁹ This mountain range is generally referred to as the Caucasus, but I use the name Caspian to maintain consistency with the English translation of the Catalan Atlas legends.

of movement and more on redemption itself, the imagery of the Caspian Mountains was intended to be ambiguous and placed the *viator* in a position to ponder the anticipated end and to view it through a subjective lens. The *viator's* understanding and interpretation of this imagery was dependent upon the beliefs they brought to their viewing of the *Atlas*. These images have predominantly been interpreted based on Christian perspectives, as the primary audience were members of the French and Aragonese courts.

Keeping with the perspective of the *Catalan Atlas's* first audience, in the Caspian Mountains, we see a variety of biblical motifs packed into the eastern-most corner of the map, woven together with a common thread of eschatological emphasis. In the densely and lavishly illustrated northeastern corner of the *Atlas*, several figures are tucked into the mountains. In the center of the sheet, on the outskirts of the mountains are two bronze statues with gold trumpets that were placed there at the request of Alexander, according to the accompanying legend.⁴⁰ Opposite them, on the inside of the mountain range, the figure identified as Alexander is placed in a position that mirrors the statues, as though looking down at his reflection. At the feet of the statues, outside the mountainous boundary, another legend describes Alexander's brush with death at the Caspian Mountains, where he would have perished had it not been for Satan, who came to his aid.⁴¹ The references in the legend to Gog and Magog, the Antichrist, and fire falling from the sky frame the interior of the mountains as a holding ground for the anticipated apocalypse. The interior of the area is divided into two halves by a strip of mountains. On one

⁴⁰ "These are made of metal and were ordered made by Alexander, a great and powerful king." Ceva, "Panel VI," <https://www.cresquesproject.net/catalan-atlas-legends/panel-vi>.

⁴¹ "The Caspian Mountains where Alexander saw trees so tall that their canopies touch the clouds. This is where he almost died had it not been for Satan who took him out of there using his arts. And with his stratagem he locked here the Tartarians God and Magog; and for them, he ordered made two metal images above described. He locked here many diverse races who don't hesitate to eat all kinds of raw meat, and from this group will come the Antichrist and their end will be caused by the fire that will fall from the sky and will confound them." Ceva, "Panel VI," <https://www.cresquesproject.net/catalan-atlas-legends/panel-vi>.

side, the ruler of Gog and Magog is identified by the legend within the enclosed mountains.⁴² He rides a black horse, wearing a red and gold crown and wielding a golden scepter. Locked within the mountains are the nations of Gog and Magog, from whom the Antichrist will emerge and bring fire that falls from the sky.

On the other side of the divide, the figure identified as Alexander is seen conversing with the winged Devil. Across from them is a crowned man holding in each hand a branch laden with coins and medallions. More pieces of gold shower the crowds on either side of the figure. The legend most commonly associated with the image reads: “Antichrist. He will be raised in Goraym of Galilea, and at the age of thirty he will start to preach in Jerusalem; contrary to the truth, he will proclaim that he is Christ, the living son of God. It is said that he will rebuild the Temple.”⁴³ Unlike the other illustrations and their accompanying legends, which are placed immediately adjacent to one another, the crowned figure is separated from this legend by the mountains below his feet. Elsewhere throughout the *Atlas* images and legends that are intended to be read together are oriented in the same direction.⁴⁴ The legend describing the Antichrist, however, is placed so that it faces directly east while the crowned figure is set on a diagonal. Though the matter of orientation and separation does not entirely sever the potential relationship between the image and legend, it is reason enough to pause and consider whether Cresques intended them to read in tandem. It should also be noted that the legend describing the nations of Gog and Magog is separated from the location of their ruler within the mountains. However, the placement of that legend at the feet of the statues and near Alexander, as well as the additional

⁴² “The Great Lord Prince of Gog and Magog. He will appear with many people in the times of the Antichrist.” Ceva, “Panel VI,” <https://www.cresquesproject.net/catalan-atlas-legends/panel-vi>.

⁴³ Ceva, “Panel VI,” <https://www.cresquesproject.net/catalan-atlas-legends/panel-vi>.

⁴⁴ For example, just outside the mountains encasing the eastern corner is an illustration of men with “strange birds,” as described by the legend immediately below their feet. For another example, see also the placement of Noah’s Ark at Mount Ararat and its legend in Figure 10.

legend identifying the ruler of Gog and Magog, provides sufficient evidence for the piecing together of the relationship between the text and image.

The identity of the crowned figure has been debated amongst scholars of the *Atlas*, resulting in a variety of understandings of how it might have been read. One possibility for the image is proposed by Sandra Sáenz-López Pérez—that perhaps this is, in fact, a Christ-like figure, though not Christ himself, intended to be reminiscent of a prophecy in the Book of Daniel about the Antichrist, though not actually the Antichrist.⁴⁵ Kogman-Appel echoes this, arguing that Cresques illustrated this figure in a manner that was intentionally ambiguous, portraying neither Christ nor Antichrist, as the visual links to other descriptions of the Antichrist are weak, as are connections to other depictions of Christ.⁴⁶ The arguments by Sáenz-López Pérez and Kogman-Appel do not, however, mean that interpretations of the figure as Antichrist should be disregarded. Instead, they imply that the image can both depict *and* not depict the Antichrist because Cresques designed the *Atlas* with the knowledge that viewers and *viatores* would bring their own beliefs and assumptions to their engagements with it. The case for this figure being *only* a depiction of the Antichrist is further weakened by the use of the future tense in the legend at the feet of the bronze statues outside the mountains and the legend identifying the ruler of Gog and Magog. On the other hand, the case is strengthened by the fact that, in both pieces of text, the Antichrist is described as one who *will* emerge, just as the nations of Gog and Magog *will* be released in the end times. Thus, the ambiguity of the relationship between the crowned figure and Antichrist legend revealed the necessity of the *viator's* own beliefs in identifying the figure.

⁴⁵ Sáenz-López Pérez, “La Representación de Gog y Magog La Imagen Del Anticristo En Las Cartas Náuticas Bejomedievales,” 263–76.

⁴⁶ Kogman-Appel, *Catalan Maps and Jewish Books*, 274. For a detailed discussion of the possible Jewish interpretations of this image, as well as the correlation of Gog and Magog to the Mongol federation, see also pages 250–253 and 260–285.

The placement of ambiguous apocalyptic imagery in the ancient mountains, juxtaposed with the depths of the unknown represented by open ocean, grounded the *viator*'s anticipation of the world's end in stone. In so doing, it anchored the eschatological in sand on the shore and in the peaks and valleys of the mountains. The contrast of strategy is instructive. Images that signify water called to mind narratives that asked the *viator* to look back and reflect on what God has done. By contrast, the solidity of the Caspian Mountains and the imagery within them prompted the *viator* to anticipate that which was to come and to prepare for it by participating in the divine order set for them. Framed thus, the end of days is as sure as the presence of the mountains themselves, but knowledge of what that end entails is as unfixed as the grains of sand on the beach. In this manner of framing, the *Catalan Atlas* reminded the *viator* that while on earth, their understanding was limited.

CONCLUSION

In the late 14th century, maps were rapidly changing—in processes of production as well as in perceptions of what they could and should do.⁴⁷ Crafted by the cartographer and scholar Elisha Ben Abraham Cresques, the *Catalan Atlas* solidified its place in the histories of geography and art for its innovative use of a compass rose. Perhaps more important, though, is the way it articulated a motif of movement that extended from the literal and physical to the metaphorical, allegorical, and arguably even anagogical. That deployment matters, for in it the *Catalan Atlas* activated a synthesis of geographical and spiritual understanding that would move its audience from one spiritual state to another. In this respect, the *Atlas* functioned as an instrument that sharpened and refined the *viator*'s spirit by setting them in motion, requiring them to navigate ambiguities and uncertainties. The *viator*—hardly a mere viewer—would have

⁴⁷ Harley, "The Map and the Development of the History of Cartography," in *The History of Cartography*, 1:26.

come away from the *Atlas* changed, their spiritual trajectory altered by a vehicle for the soul itself.

As a vehicle, the *Catalan Atlas* reminded its *viatores* that while the universe and the afterlife were thought to be infinite, the earth and one's time on it were finite, limited, and plagued by uncertainty. And yet, the original audience for this object would have been keenly aware that the relationship between body and soul is temporal. As the human body returned to dust, the soul would return to its home in the cosmos. Thus, life on earth was to be lived in such a way that a finite human life flowed seamlessly into the afterlife. Facing the *Catalan Atlas*, one would have found its six sheets a continual reminder to move through this world accordingly.

By way of closing, I would add that the *viator* who grapples with those sheets has changed, but the challenge of the *Catalan Atlas* remains remarkably consistent. Its challenges and affordances call on us to situate ourselves within the broader context of the world in which we live and its place in the framework of the universe. Repeatedly alluding to things of greater importance, the *Catalan Atlas* repeatedly enjoins us to consider something larger than the individual self. It provides continual reminders of our impending deaths both by referencing that phenomenon and by framing us as *viatores*, whose own trajectory is ultimately the same as those of the people who move and struggle and travel across its surface. In this way, the *Catalan Atlas* encourages us to become the fullest versions of ourselves and welcome the abilities and insights each of us has to offer. Perhaps, then, *viator's* journey through the *Catalan Atlas* is just this—a quest for the life that makes earthly life worth living.

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Figure 8: *Catalan Atlas*, Paris, Bibliothèque nationale de France, cod. Esp 30, attributed to Elisha Ben Abraham Cresques, Majorca, c. 1375. Vellum mounted on wood panels. 64.5cm x 50cm. Sheet 5, detail: Noah's Ark at Mount Ararat.

Figure 9: *Catalan Atlas*, Paris, Bibliothèque nationale de France, cod. Esp 30, attributed to Elisha Ben Abraham Cresques, Majorca, c. 1375. Vellum mounted on wood panels. 64.5cm x 50cm. Sheet 4, detail: Red Sea.

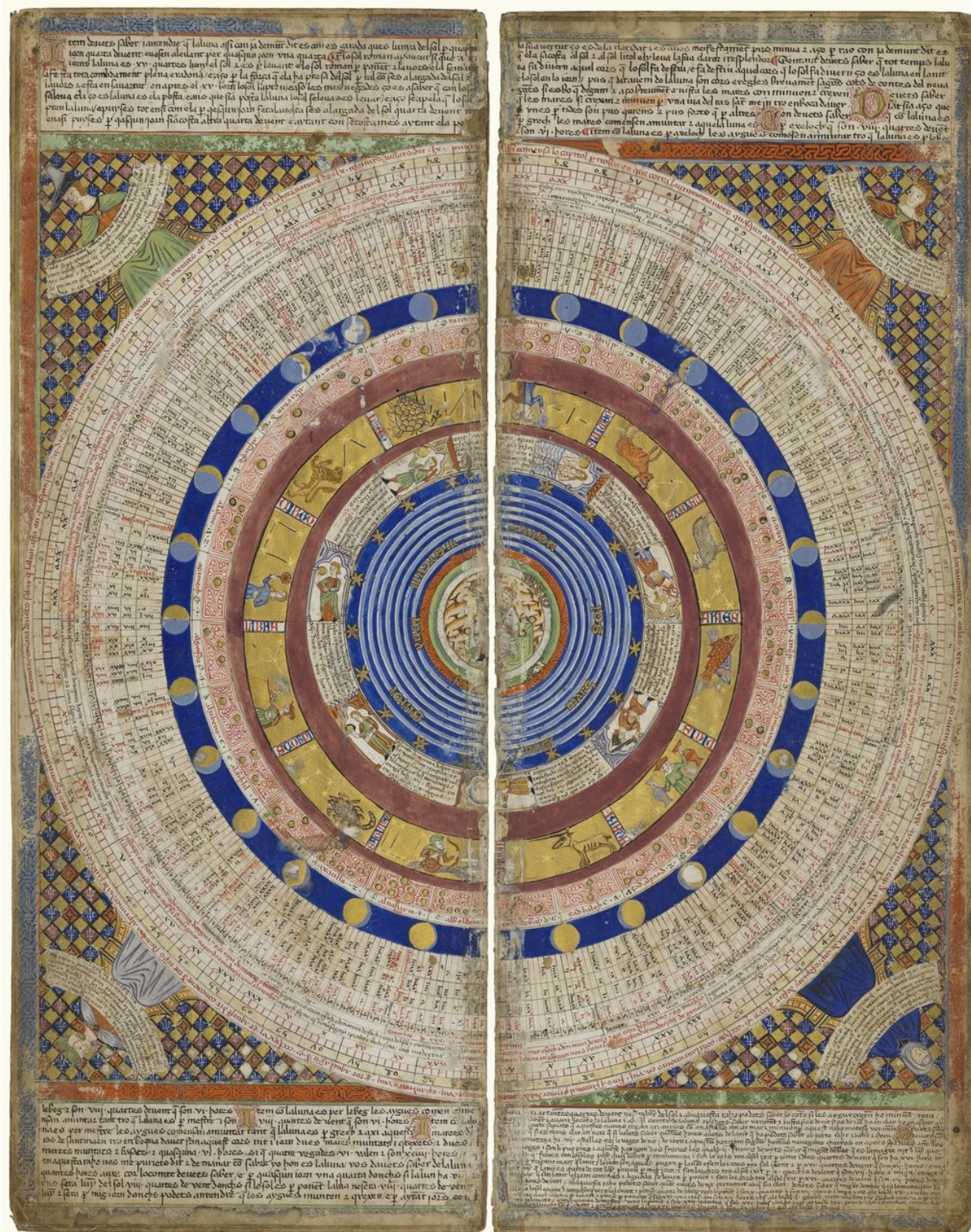
Figure 10: *Catalan Atlas*, Paris, Bibliothèque nationale de France, cod. Esp 30, attributed to Elisha Ben Abraham Cresques, Majorca, c. 1375. Vellum mounted on wood panels. 64.5cm x 50cm. Sheet 5, detail: Lop Desert.

Figure 11: *Catalan Atlas*, Paris, Bibliothèque nationale de France, cod. Esp 30, attributed to Elisha Ben Abraham Cresques, Majorca, c. 1375. Vellum mounted on wood panels. 64.5cm x 50cm. Sheet 6, detail: Caspian Mountains.



Source gallica.bnf.fr / Bibliothèque nationale de France. Département des Manuscrits. Espagnol 30

Figure 1a



Source gallica.bnf.fr / Bibliothèque nationale de France. Département des Manuscrits. Espagnol 30

Figure 1b



Figure 1c



Figure 1d

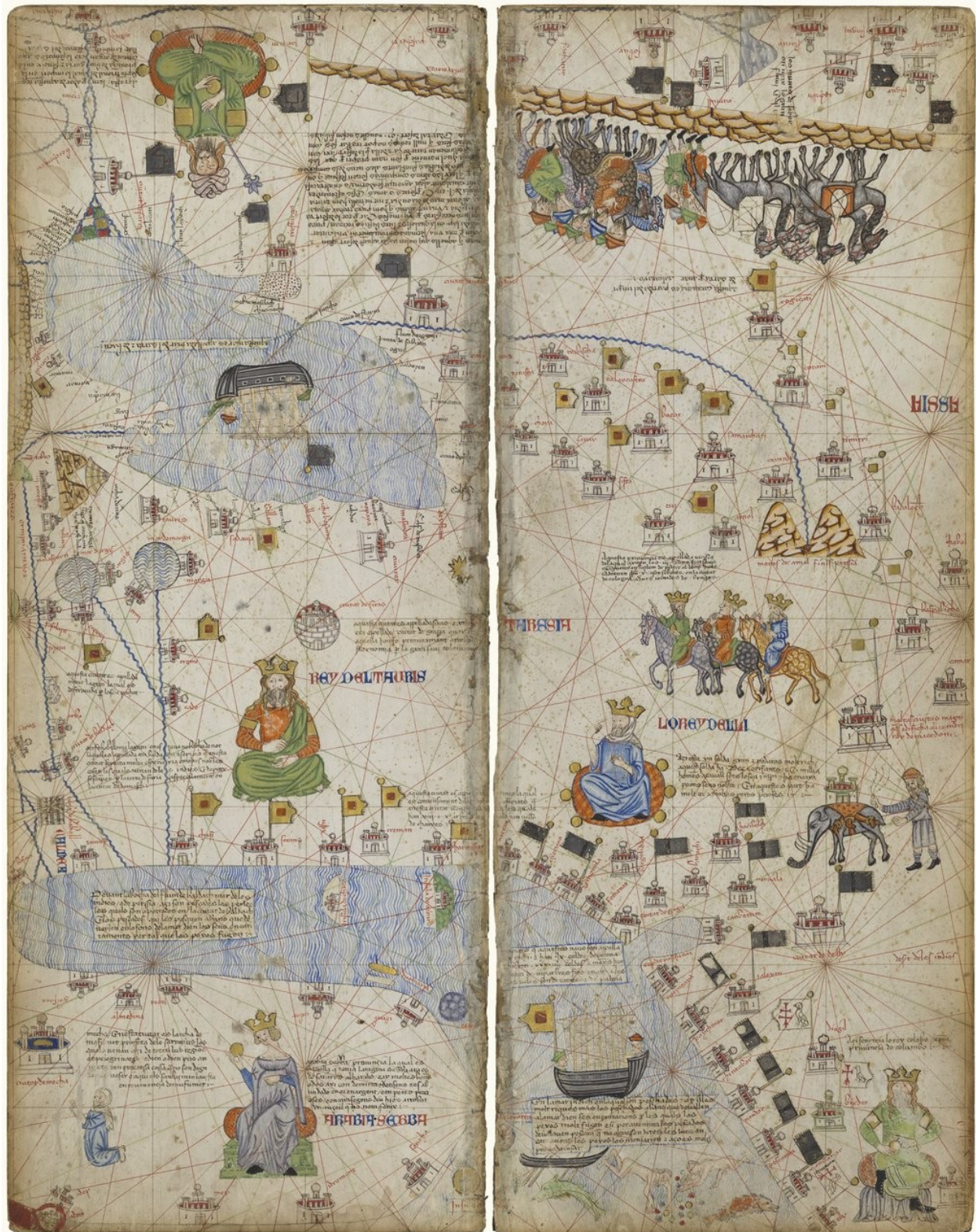


Figure 1e

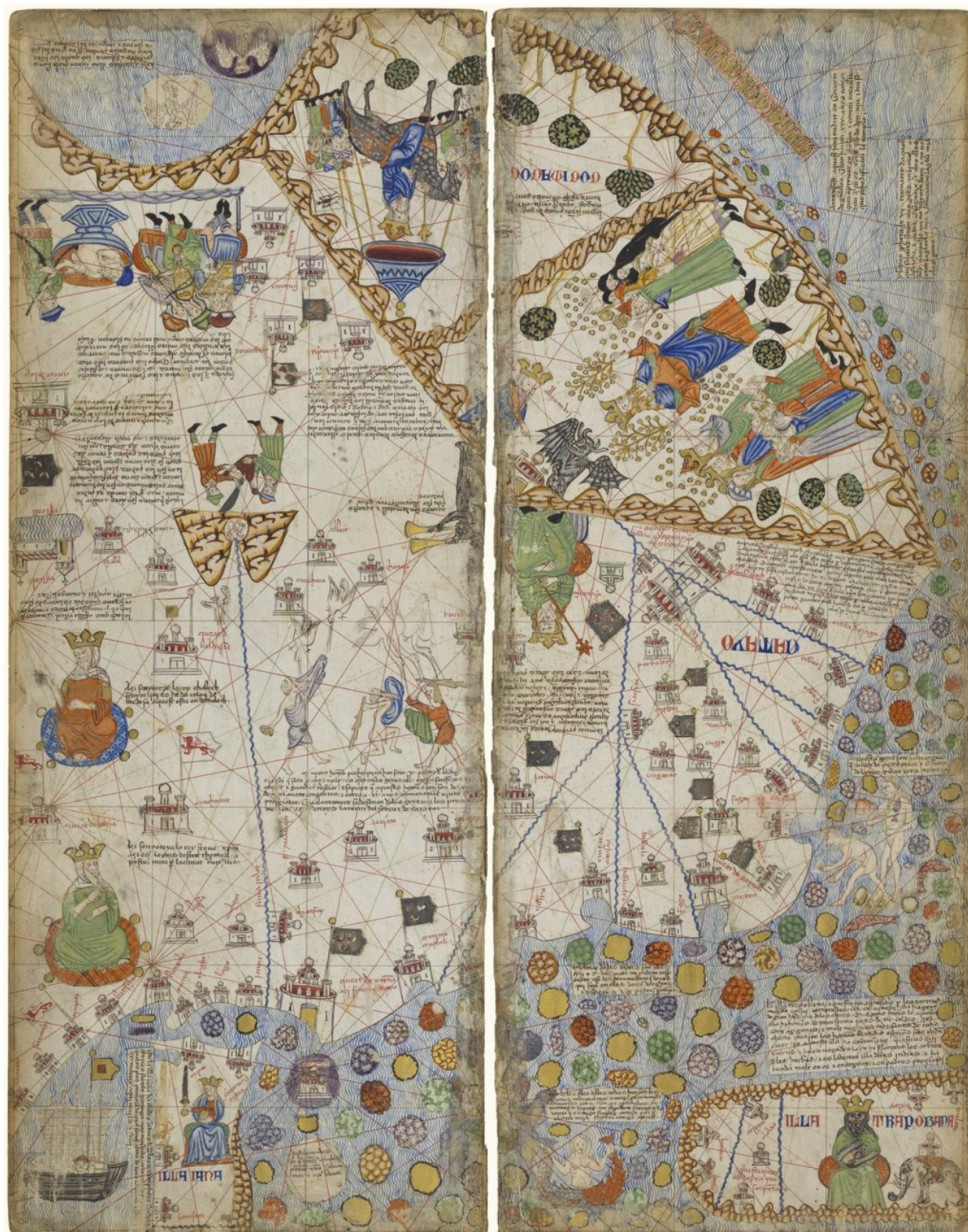


Figure 1f



Figure 2



Figure 3



Figure 4



Figure 5

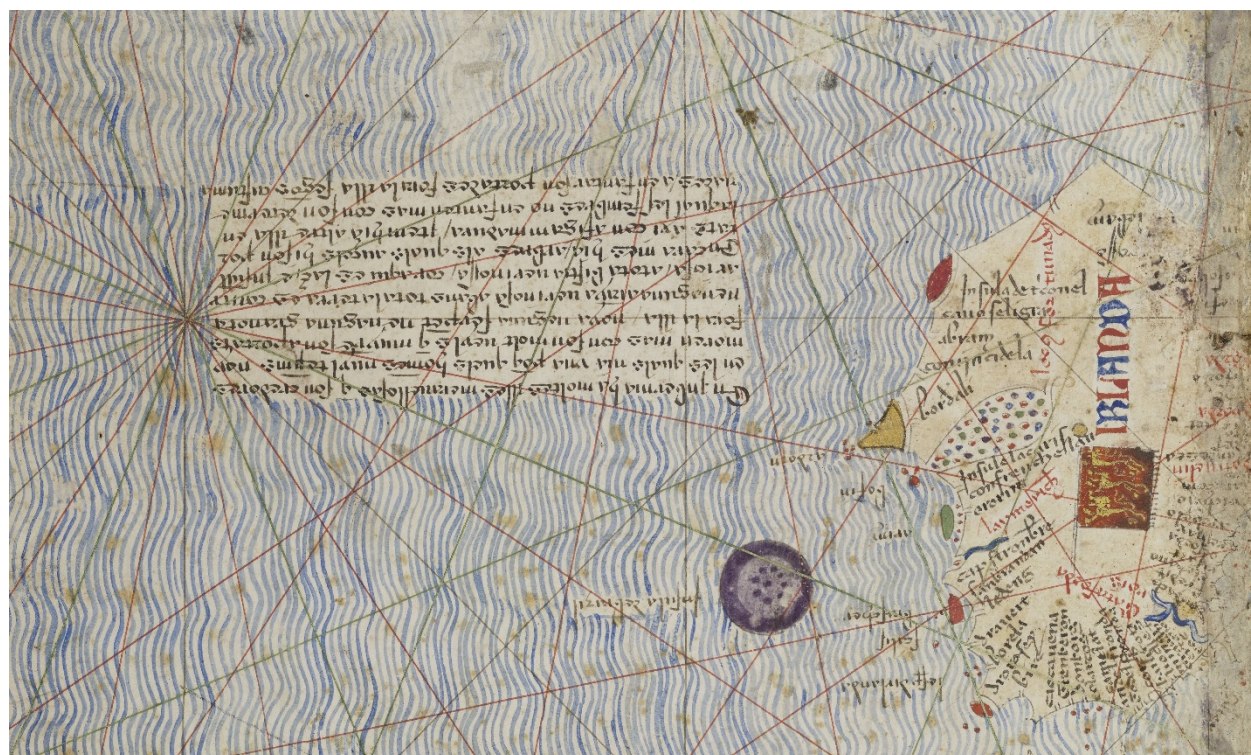


Figure 6



Figure 7



Figure 10



Figure 11